

Cyclic Group

- If there exists an element $a \in G$ for some group $(G, *)$ such that every element of G can be written as some power of a , that is a^n for some integer n . Then the group $(G, *)$ is said to be cyclic group and the element a is called a generator of the group.

Properties:

- 1) Every cyclic group is abelian.
- 2) If a is a generator of a cyclic group $(G, *)$, then a^{-1} is also a generator.
- 3) If a is a generator of a cyclic group of order n , then $a^n = e$.
- 4) If $(G, *)$ is a finite cyclic group of order n generated by a , then a^m is also generator of $(G, *)$ iff m and n are relatively prime.

Example: $(\{1, -1, i, -i\}, \times)$ is a cyclic group with i as a generator

$$i^1 = i, \quad i^2 = -1, \quad i^3 = i^2 \times i = -1 \times i = -i, \quad i^4 = 1$$

So, I have generated all the four numbers by using the element i , so this is a cyclic group and i is a generator. And you know that the inverse of i is $-i$ so $-i$ is also a generator of the group.

Let's see

$$(-i)^1 = -i, \quad (-i)^2 = -1, \quad (-i)^3 = i, \quad (-i)^4 = 1$$

Let see can 1 be the generator?

$$1^1 = 1, \quad 1^2 = 1, \quad 1^3 = 1 \text{ and so on...}$$

So only 1 element is being generated so it cannot be a

generator.

Let see -1 ?

$(-1)^1 = -1$, $(-1)^2 = 1$, $(-1)^3 = -1$, $(-1)^4 = 1$, So (-1) cannot be a generator.

Hint: See here $(-1)^2 = 1$ where 1 is the identity.
 $O(G) = 4$

So if you are getting the identity element by a power less the order then it cannot be a generator.

Example: Show that $G = (\{1, 2, 4, 5, 7, 8\}, \times_9)$ is cyclic.

How many Generators are there?

Sol Identity element = 1.

1 cannot be a generator. why?

Since Identity element cannot be a generator.

Let's check 2.

$$2^1 = 2 \equiv 2 \pmod{9}, \quad 2^2 = 4 \equiv 4 \pmod{9}$$

$$2^3 = 8 \equiv 8 \pmod{9}, \quad 2^4 = 16 \equiv 7 \pmod{9}$$

$$2^5 = 32 \equiv 5 \pmod{9}, \quad 2^6 = 64 \equiv 1 \pmod{9}$$

6th power give me 1 (identity) So 2 is a generator.

If 2 is a generator then inverse of 2 will be a generator.

Inverse of 2 is 5. So you can also verify

$$5^1 = 5 \pmod{9}, \quad 5^2 = 7 \pmod{9}, \quad 5^3 = 8 \pmod{9}, \quad 5^4 = 4 \pmod{9}$$

$$5^5 = 2 \pmod{9}, \quad 5^6 = 1 \pmod{9}.$$

Let's try 4?

$$4^1 = 4, \quad 4^2 = 16 \equiv 7 \pmod{9}, \quad 4^3 = 64 \equiv 1 \pmod{9}$$

So 4 cannot be a generator.

If 4 cannot be a generator then its inverse 7 can also not be a generator.

	1	2	4	5	7	8
1	1	2	4	5	7	8
2	2	4	8	1	5	7
4	4	8	7	2	1	5
5	5	1	2	7	8	4
7	7	5	1	8	4	2
8	8	7	5	4	2	1

Similarly if I am taking 8?

$$8^1 = 8, \quad 8^2 = 64 \equiv 1 \pmod{9}$$

So once again 8 cannot be a generator.

So only 2 generators are 2 & 5.

Ex. Show that the Group $(G = \{0, 1, 2, 3, 4\}, +_5)$ is a cyclic Group. What are the Generators.

Sol.

Ex. How many generators are there of a cyclic group of order 8? What are these?

Sol. We know that if $(G, *)$ is a finite group of order n with a as a generator then a^m is also a generator iff m & n are relatively prime.

$n = 8$, 1, 3, 5, 7 are relatively prime to 8.

So there will be 4 generators.

The generators (in terms of the presumed generator a) are

$$a^1 = a, a^3, a^5, a^7.$$

Tests for Subgroup

Two Step Test: A non-empty subset H of a group G is a subgroup of G iff.

- i) $a * b \in H$ for $a, b \in H$
- ii) $a^{-1} \in H$ for every $a \in H$.

One Step Test: The necessary and sufficient condition for a non-empty subset H of $(G, *)$ to be a subgroup is $a * b^{-1} \in H$ for $a, b \in H$ where b^{-1} is inverse of b in G .

Ex Show that the set $H = \{a+ib \in \mathbb{C} \mid a^2+b^2=1\}$ is a subgroup of $(\mathbb{C}, \cdot)$ where $\cdot$ is the multiplication of complex numbers.

Sol. Using the two step Test:

Let $a+ib, c+id \in \mathbb{C}$ s.t. $a^2+b^2=1$ and $c^2+d^2=1$

Step 1. $(a+ib)(c+id)$

$$= (ac-bd) + i(ad+bc)$$

checking

$$\begin{aligned} & (ac-bd)^2 + (ad+bc)^2 \\ &= a^2c^2 + b^2d^2 - 2abcd + a^2d^2 + b^2c^2 + 2abcd \\ &= a^2c^2 + b^2d^2 + a^2d^2 + b^2c^2 \\ &= a^2(c^2+d^2) + b^2(c^2+d^2) \\ &= a^2+b^2 = 1. \end{aligned}$$

Step 1 verified.

So, H is closed under the operation.

Step 2. As operation is multiplication identity element = 1.

Let $x^{-1} \in H$ is inverse of $x \in H$ and $x = a+ib, a^2+b^2=1$

$$\text{Then } xx^{-1} = 1 \Rightarrow x^{-1} = \frac{1}{x} = \frac{1}{a+ib} = \frac{a-ib}{a^2+b^2} \quad a^2+b^2 \neq 0$$

$$= a-ib \quad [\text{as } a^2+b^2=1]$$

So for $x = a+ib \in H$ then

$$x^{-1} = a-ib \in H$$

$$[a^2+(-b)^2 = a^2+b^2=1]$$

So Step 2 verified.

Permutation Group or Symmetric Group

Let S be a set of n element. A one-one onto mapping from S to S is called permutation on S .

The set of all permutation is denoted by S_n .

Representation of Permutation:

$$S = \{a_1, a_2, \dots, a_n\}$$

Let $f \in S_n$ and $f(a_1) = b_1, f(a_2) = b_2, \dots, f(a_n) = b_n$ [$b_i \in S$]

$$f = \begin{pmatrix} a_1 & a_2 & a_3 & \dots & a_n \\ b_1 & b_2 & b_3 & \dots & b_n \end{pmatrix}$$

Ex. $S = \{1, 2, 3\}$

$$f = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, g = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, h = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

Total 3! ways. $S_n = o(S_n) = 6$

The set of permutation (mappings) is called permutation group or symmetric group of degree n denoted by S_n .

$$o(S_n) = n!$$

Identity Permutation:

$$I = \begin{pmatrix} a_1 & a_2 & a_3 & \dots & a_n \\ a_1 & a_2 & a_3 & \dots & a_n \end{pmatrix}$$

$$I(a_1) = a_1, I(a_2) = a_2, \dots, I(a_n) = a_n.$$

Ex. $S = \{1, 2, 3, 4\}$

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

Equality of Two Permutation:

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad g = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix}$$

$$f(1) = 2, g(1) = 2, f(2) = 3, g(2) = 3, f(3) = 4, g(3) = 4, \\ f(4) = 1, g(4) = 1 \text{ So } f = g.$$

Product of Two Permutations:

Let $f = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{pmatrix}$ and $g = \begin{pmatrix} b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}$

Then $fg = fog = \begin{pmatrix} a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \end{pmatrix}$

Domain of g = Co-domain of f .

Ex $f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix}$, $g = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$

$$fg = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad gf = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}$$

$$fg \neq gf$$

So product of permutation is not commutative.

Inverse Permutation:

Permutation is one-one, onto. Hence it is invertible always.

Ex $f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix}$

Let g be the inverse permutation of f

$$g = \begin{pmatrix} 1 & 2 & 3 & 4 \\ x & y & z & w \end{pmatrix}$$

$$fg = I \Rightarrow \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ x & y & z & w \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

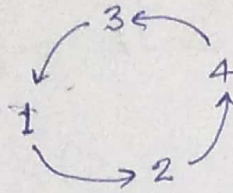
$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ y & w & z & x \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

Comparing, we get $y=1, w=2, z=3, x=4$

So $g = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix}$

Cyclic Permutation:

Ex. Permutation $f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 5 & 1 & 6 & 3 \end{pmatrix}$ is cyclic.



$$\begin{array}{c} (1 \ 2 \ 4 \ 3) = f \\ \hline \text{cycle} \end{array}$$

Ex. Cycle $(1 \ 3 \ 4 \ 2 \ 6)$ represents a permutation of degree 9 on a set S consisting of the elements 1, 2, 3, 4, 5, 6, 7, 8, 9. Then the permutation will be:

$$(1 \ 3 \ 4 \ 2 \ 6) = \begin{pmatrix} 1 & 3 & 4 & 2 & 6 & 5 & 7 & 8 & 9 \\ 3 & 4 & 2 & 6 & 1 & 5 & 7 & 8 & 9 \end{pmatrix}$$

Length of cycle:

No. of elements in cycle = Length of cycle

Ex. Permutation $g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 4 & 6 & 5 & 1 \end{pmatrix}$ is cyclic.

$$(1 \ 2 \ 3 \ 4 \ 6) = g$$

$$\text{Length} = 5$$

Ex. Permutation $h = \begin{pmatrix} 1 & 2 & 4 & 3 & 5 & 6 \\ 2 & 1 & 3 & 4 & 5 & 6 \end{pmatrix}$ is not cyclic.

$1 \rightarrow 2$ It's not form a cyclic so its not a cyclic.

Disjoint Cycles:

Two cycles are said to be disjoint if they have no symbols in common.

Ex. Cycles $(1 \ 3 \ 5)$ and $(2 \ 6 \ 8 \ 9)$ are disjoint cycles. while

Cycles $(1 \ 3 \ 4)$ and $(2 \ 3 \ 5 \ 6)$ are not disjoint.

Ex. Let $f = (1\ 2\ 3)$ and $g = (4\ 5)$ represents two permutations on 5 symbols 1, 2, 3, 4, 5. Then:

$$fg = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{pmatrix} \cdot \begin{pmatrix} 4 & 5 & 1 & 2 & 3 \\ 5 & 4 & 1 & 2 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$$

$$gf = \begin{pmatrix} 4 & 5 & 1 & 2 & 3 \\ 5 & 4 & 1 & 2 & 3 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix} \quad \boxed{fg = gf}$$

NOTE: Disjoint cycles is commutative.

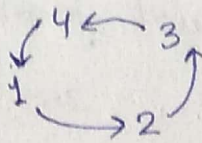
* Every permutation can be expressed as a product of disjoint cycles.

Ex. $f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 3 & 1 & 4 & 8 & 6 & 9 & 7 & 5 \end{pmatrix}$

$(4) (6) (1\ 2\ 3) (5\ 8\ 7\ 9)$

Properties of Cycle:

Cycle $\rightarrow (1\ 2\ 3\ 4) = (2\ 3\ 4\ 1) = (3\ 4\ 1\ 2) = (4\ 1\ 2\ 3)$



Result (1): A cycle does not change by changing the places of its elements provided their cyclic order.

(2) A cycle of length 1 is called Identity Permutation.

Transposition:

A cycle of length two is called Transposition.

Ex. $(2, 3)$ is transposition.

Multiplication of Cycles:

Ex. If cycles $(1\ 2\ 3)$ and $(5\ 6\ 4\ 1)$ represents permutations of degree 6 on six symbols 1, 2, 3, 4, 5, 6. Then:-

$$\begin{aligned}(1\ 2\ 3) \cdot (5\ 6\ 4\ 1) &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 1 & 4 & 5 & 6 \end{pmatrix} \cdot \begin{pmatrix} 5 & 6 & 4 & 1 & 2 & 3 \\ 6 & 4 & 1 & 5 & 2 & 3 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 5 & 1 & 6 & 4 \end{pmatrix} \\ &= (1\ 2\ 3\ 5\ 6\ 4)\end{aligned}$$

Ex. $(1) \cdot (2\ 3\ 4) \cdot (6) = (2\ 3\ 4)$

$\downarrow$ Identity $\downarrow$ Identity

Even and Odd Permutation:

$$f = (1\ 2\ 3\ 4\ 5)$$

$$\text{Transpositions} = (1\ 2)\ (1\ 3)\ (1\ 4)\ (1\ 5)$$

$$g = (1\ 2\ 3\ 4\ 5\ 6\ 7)$$

$$\tau = (1\ 2)\ (1\ 3)\ (1\ 4)\ (1\ 5)\ (1\ 6)\ (1\ 7)$$

* A cycle of length n can be expressed as the product of $(n-1)$ transpositions.

* A permutation is said to be even if it can be expressed as a product of even number of transpositions, otherwise it is odd.

Ex: $f = (1\ 2\ 3)\ (1\ 2)$

$$f = (1\ 2)\ (1\ 3)\ (1\ 2)$$

Transpositions = 3 (odd) $\therefore f$ is odd permutation.

Q Determine which of the following are even permutations:

a) $f = (12345)(123)(45)$

b) $g = (12)(13)(14)$

Sol. b, $g = (12)(13)(14)$

So ^{odd} g is odd permutation.

a) $f = (12)(13)(14)(15)(12)(13)(45)$

Transpositions = 7 (odd)

$\therefore f$ is odd permutation.

Properties:

1) Identity permutation is always even permutation.

2) Product of two even permutations is an even permutation.

3) Product of two odd permutations is an even permutation.

4) Product of an even permutation and an odd permutation is an odd permutation.

5) Total no. of even permutations of degree n
 $= \frac{n!}{2}$

6) Total no. of odd permutations of degree $n = \frac{n!}{2}$

Homomorphism

Defⁿ: Let G and G' be any two groups with binary operation $\circ$ and $\circ'$ respectively. Then a mapping $f: G \rightarrow G'$ said to be homomorphism if

$$\forall a, b \in G, \quad f(a \circ b) = f(a) \circ' f(b)$$

Ex: $G = \{1, -1, i, -i\}$ w.r.t. multiplication & I be group of all integers under addition.

Prove that $f: I \rightarrow G$ is a homomorphism where $f(n) = i^n \forall n \in I$

Sol. $f(m) = i^m$ & $f(n) = i^n \forall n, m \in I$

$$f(m+n) = i^{m+n} = i^m \cdot i^n$$

$$f(m+n) = f(m) \cdot f(n)$$

So $f: I \rightarrow G$ is a homomorphism.

Ex. Cyclic group $G = \{a, a^2, a^3, \dots, a^{12} = e\}$, and its subgroup $G' = \{a^2, a^4, a^6, \dots, a^{12}\}$, where $f(a^n) = a^{2n}$. $f: G \rightarrow G'$

Prove that f is homomorphism.

Sol Let $a^n, a^m \in G$ $f(a^n) = a^{2n}$ & $f(a^m) = a^{2m}$

$$\begin{aligned} f(a^n \cdot a^m) &= f(a^{n+m}) \\ &= a^{2(n+m)} = a^{2n} \cdot a^{2m} \\ &= f(a^m) \cdot f(a^n) \end{aligned}$$

Isomorphism of Groups

Let $(G, \circ)$ and $(G', \circ')$ be two gps, then a function $f: G \rightarrow G'$ is called an Isomorphism if:

1) f is a homomorphism.

2) f is a one-one onto mapping.

Then G' is called isomorphic image of G and written as:

$$G \cong G'$$

Ex. Let R be the additive group of real numbers and R^+ be the multiplicative group of positive real numbers and $f: R \rightarrow R^+$ defined by

$$f(x) = e^x \quad \forall x \in R$$

then show that $R \cong R^+$.

Sol. 1) f is homomorphism:

Let $x, y \in R$ then $(R, +)$

$$\begin{aligned} f(x+y) &= e^{x+y} = e^x \cdot e^y \\ &= f(x) \cdot f(y) \end{aligned}$$

$$f(x), f(y) \in (R^+, \cdot)$$

$\therefore f$ is homomorphism.

2) f is one-one mapping.

Let $x, y \in R$ and

$$\begin{aligned} \text{Let } f(x) &= f(y) \\ e^x &= e^y \end{aligned}$$

Now taking $\log$ on both sides, we get

$$x = y$$

$\therefore f$ is one-one mapping.

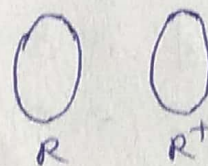
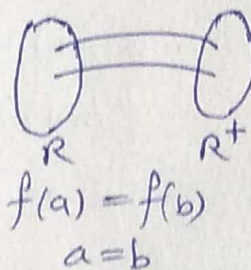
f is onto mapping.

Let $z \in R^+$ be any arbitrary element, then there exists at least one element $\log z \in R$ s.t.

$$\begin{array}{ccc} f(\log z) &= e^{\log z} &= z \\ \downarrow & & \downarrow \\ R & & R^+ \end{array}$$

$\therefore f$ is onto mapping.

Hence, $R \cong R^+$ (f is isomorphism)



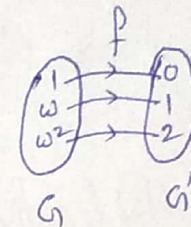
Ex. Let $G = \{1, \omega, \omega^2\}$ w.r.t. multiplication & $G' = \{0, 1, 2\}$ w.r.t. addition (mod 3). Show that $G \cong G'$.

Sol

$\cdot$	1	ω	ω^2
1	1	ω	ω^2
ω	ω	ω^2	1
ω^2	ω^2	1	ω

$+_3$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$$\begin{aligned} 1 &\rightarrow 0 & \omega &\rightarrow 1 & \omega^2 &\rightarrow 2 \\ f(1) &= 0 & f(\omega) &= 1 & f(\omega^2) &= 2 \end{aligned}$$



- 1) f is one-one mapping.
- 2) f is also onto mapping.
- 3) f is homomorphism.

$$\begin{aligned} \omega, \omega^2 &\in (G, \cdot) \\ f(\omega \cdot \omega^2) &= f(\omega^3) \quad \because \omega^3 = 1 \\ &= f(1) = 0 \\ &= 1 +_{\text{mod } 3} 2 \\ &= f(\omega) +_3 f(\omega^2) \end{aligned}$$

$\therefore f$ is homomorphism.

Hence $G \cong \underline{G'}$

COSETS

Let H be a subgroup of a group G , whose composition have been denoted multiplicatively.

Let $a \in G$, then

$$Ha = \{ha : h \in H\} \rightarrow \text{Right Coset}$$

$$\& aH = \{ah : h \in H\} \rightarrow \text{Left Coset}$$

are known as Right Cosets (Ha) and Left Coset (aH) respectively.

Ex. 1. Multiplicative Group $G = \{1, -1, i, -i\}$

$$\text{Subgp } H = \{1, -1\}$$

$$H \cdot 1 = \{1, -1\} = H \text{ and } 1 \cdot H = \{1, -1\}$$

$$H \cdot (-1) = \{-1, 1\} = H \quad -1 \cdot H = \{-1, 1\}$$

$$H \cdot i = \{i, -i\} \quad i \cdot H = \{i, -i\}$$

$$H \cdot -i = \{-i, i\} \quad -i \cdot H = \{-i, i\}$$

Properties:

- 1) $H \cdot e = e \cdot H = H$
- 2) $H \cdot h = H \Leftrightarrow h \in H$
- 3) $Ha \cap Hb = \emptyset$ OR $Ha = Hb$
- 4) $a \in Ha$
- 5) $|Ha| = |H| = |aH|$
- 6) In general $aH \neq Ha$
- 7) $Ha = aH \Leftrightarrow G$ is abelian.
- 8) Ha is subgroup of $G \Leftrightarrow a \in H$
- 9) G is Union of all right cosets of H in G .

* If the composition in the group G has been denoted additively, then

$$\text{Right Coset: } H+a = \{h+a: h \in H\}$$

$$\& \text{ Left Coset: } a+H = \{a+h: h \in H\}$$

Ex. $(\mathbb{Z}, +)$: Additive group of Integers:

$$H = (3\mathbb{Z}, +)$$

$$\therefore H = \{\dots, -9, -6, -3, 0, 3, 6, 9, \dots\}$$

$$H = H+0 = \{\dots, -9, -6, -3, 0, 3, 6, 9, \dots\}$$

$$H+1 = \{\dots, -8, -5, -2, 1, 4, 7, 10, \dots\}$$

$$H+2 = \{\dots, -7, -4, -1, 2, 5, 8, 11, \dots\}$$

$$H = H+3 = \{\dots, -6, -3, 0, 3, 6, 9, \dots\}$$

$$H+4 = H+1 \quad | \quad H+5 = H+2$$

$$* \quad H \cup H+1 \cup H+2 + \dots = \mathbb{Z}$$

LAGRANGE'S THEOREM

"The order of each subgroup of a finite group is a divisor (factor) of the order of the group." i.e. $o(H) \mid o(G)$

Example: Multiplicative group $G = \{1, -1, i, -i\}$
and Subgroup $H = \{1, -1\}$

$$o(G) = 4 \quad \& \quad o(H) = 2, \quad \text{Clearly, } o(H) \mid o(G).$$

$$* \quad \frac{o(G)}{o(H)} = \frac{4}{2} = 2 \rightarrow \text{Total number of right (Left) distinct cosets of } H \text{ in } G.$$

$$\boxed{\frac{o(G)}{o(H)} = [G:H]}$$

This is called Index $\leftarrow [G:H]$
of subgroup H in G

* The order of every element of a finite group G is divisor of the order of the group. $\downarrow a \in G$ then
 $o(a) \mid o(G)$

Proof: Let G be a group of a finite order n i.e.
 $|G| = n$

Let H be a subgroup of G of a finite order m
 i.e. $|H| = m$

Suppose $H = \{h_1, h_2, \dots, h_m\}$

Let $a \in G$. Then

$$Ha = \{h_1a, h_2a, \dots, h_ma\}$$

Ha has m distinct members.

Let us suppose that $h_ia = h_ja$ ($i \neq j$)

$$\Rightarrow h_i = h_j \quad [\text{by Cancellation Law}]$$

But it's not possible since order of H is m , if it's possible then order of H will be less than m . (It's a contradiction)

Therefore, each right coset of H in G has m distinct members.

Since G is finite group, the number of distinct right cosets of H in G will be finite, say equal to k .

The union of these k distinct right cosets of H in G is equal to G , then

$$G = Ha_1 \cup Ha_2 \cup \dots \cup Ha_k \quad [Ha_i \cap Ha_j = \emptyset]$$

$\Rightarrow$ The number of elements in G

= The no. of elements in Ha_1 + the no. of elements in Ha_2 + $\dots$ + the no. of elements in Ha_k

$$\Rightarrow |G| = km$$

$$n = km \quad \Rightarrow \quad k = \frac{n}{m}$$

$\Rightarrow m$ is a divisor of n

$\Rightarrow |H|$ is divisor of $|G|$.